

Quasi-Synchronization of Delayed Memristive Neural Networks via Region-Partitioning-Dependent Intermittent Control

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Abstract—This paper aims at investigating the master-slave quasi-synchronization of delayed memristive neural networks (MNNs) by proposing a region-partitioning-dependent intermittent control. The proposed method is described by three partitions of non-negative real region and an auxiliary positive definite function. Whether the control input is imposed on the slave system or not is decided by the dynamical relationships among the three subregions and the auxiliary function. From these ingredients, several succinct criteria with the associated co-design procedure are presented such that the synchronization error converges to a predetermined level. The proposed intermittent control scheme is also applied to the event-triggered control, and an intermittent event-triggered mechanism is devised to investigate the quasi-synchronization of MNNs correspondingly. Such mechanism eliminates the events in rest time, and then it reduces the amount of samplings. Finally, two illustrative examples are presented to verify the effectiveness of our theoretical results.

Index Terms—Intermittent control, intermittent event-triggered mechanism (IETM), memristive neural networks (MNNs), quasi-synchronization analysis, time-varying delays.

I. INTRODUCTION

IN THE last few decades, the research on delayed neural networks has received considerable attention due to their applications in various areas [1]–[3]. It has been shown that the chosen time-delays and connection weights of neural networks are likely to lead to complicated behaviors even with strange chaotic attractors [4], [5]. The synchronization control of these complicated behaviors can find many potential applications, such as the secure communications, biological systems, information science, image encryption, to name a few [7], [8]. With a steadily growing interest, an extraordinary volume of literatures are in the existence on the synchronization

of delayed neural networks, see [5], [9]–[16], and the references therein. Meanwhile, various control schemes have been developed. These methods include but are not limited to the state/output feedback control [5], [6], adaptive control [9]–[12], sampled-data control [13]–[15], impulsive control [16]–[18], and intermittent control. The analogues of synchronization include the tracking problem, filtering, and consensus of multiagent systems, etc. [19]–[22].

Since the seminal work of [23], an important type of neural networks, which is called memristive neural networks (MNNs), has provoked considerable attention. The MNNs are formed by replacing the resistors in traditional recurrent neural networks with memristors. The research on MNNs will give us inspirations on how to apply them in real applications and further insight into the associative memories, neural learning circuits, large-scale neuromorphic systems, and so on [24]–[28]. Unlike the traditional recurrent neural networks, the connection weights of MNNs are changed according to the state dynamics, so such networks are state-dependent switching systems [29]–[31]. The switching property of MNNs results in that their synchronization is much more intricate and complex than the resistor-based ones. The main reason is that the memristor-based master and slave systems may switch asynchronously owing to their different states. In other words, the synaptic weights of master and slave systems may be nonidentical before achieving synchronization [32]–[35]. In addition, MNNs belong to the class of differential equations with discontinuous right-hand sides. It is reminiscent of the analysis results in [36]–[38] that the classical feedback controller is tough to achieve the complete synchronization for discontinuous systems even in the absence of parameter mismatch. These results inspire us to explore the quasi-synchronization of MNNs in this paper. Different from the complete synchronization, quasi-synchronization only demands that the state error is controlled within a reasonable region. In fact, compared with complete synchronization, quasi-synchronization is more reasonable and practical, due to the unavoidability of perturbation in operation [5], [16], [36].

On the other hand, intermittent control is an effective strategy to achieve synchronization. In such control, the time is composed of work time (or control time) where the control input is imposed, and rest time where any control is not added. Compared with the continuous control scheme, it is a more economic choice and can simulate the real world better [39]. Meanwhile, intermittent control can increase the

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difficulty when capturing the information by sending a signal, then it is also more safe in secure communications. In the pioneering work [39] on synchronization of delayed chaotic systems with parameter mismatches, the authors first proposed the intermittent feedback control to have the systems reaching quasi-synchronization. This original method is innovative and efficient. Periodically intermittent control is one of the most common intermittent types, where the work time and rest time are periodic and clock-dependent [30], [39]–[42]. However, the requirement of periodicity is quite restricted and unnecessary in applications. For example, the generation of wind power is typically aperiodically intermittent. In [43] and [44], the aperiodically intermittent pinning control is proposed to investigate the synchronization of complex networks. Although this alternative strategy overcomes the limitations of periodically intermittent control, the proportion of work time, i.e., the index

$$\delta_w = \frac{\text{work time}}{\text{work time} + \text{rest time}} \times 100\%$$

is limited by a certain lower bound. Moreover, in the presence of time-delays, the proportion will be rapidly increased [44]. Very recently, the state-dependent intermittent control scheme is proposed in [45]. This scheme divides the state space into three subspaces in advance, and it sets the work time and rest time online according to the relationships among the system states and three subspaces. An attractive characteristic of state-dependent intermittent control scheme is that the index δ_w is not related directly to the system parameters and has no a certain lower bound. Thus, it is more competitive than most of the existing ones. However, the method in [45] only applies to the complete exponential stability of closed-loop systems, thus it cannot be duplicated directly to investigate quasi-synchronization of MNNs. It also should be stressed that the scheme in [45] is restricted to the systems without time-delays, which remains much room for improvement.

Motivated by the foregoing discussion, this paper provides insights into quasi-synchronization of MNNs by proposing an alternative intermittent control scheme. The main contributions are summarized as follows.

- 1) The non-negative real region is first divided into three subregions. Meanwhile, an auxiliary positive definite function is introduced to govern the work time and rest time by measuring its relationships with the three subregions online, which motivates the proposed name of region-partitioning-dependent intermittent control. Such intermittent control belongs to the aperiodic type. Since the auxiliary function contains the information of time delays, the proposed scheme is suitable to analyze the quasi-synchronization of delayed MNNs. Under the framework of our scheme, some sufficient conditions are formulated to ensure the boundedness of synchronization error. Whereafter, an associated co-design procedure is provided such that the synchronization error can be controlled in a predetermined level.
- 2) Note that a controller is usually implemented in a digital way. In view of this, only the discrete-time state measurements are transmitted from the sensor to controller through a communication network. Motivated

by this fact, we incorporate the event-triggered control into our intermittent control scheme, and an intermittent event-triggered mechanism (IETM) is devised originally to investigate the quasi-synchronization of MNNs. Especially, since the proposed IETM only focuses on the events in the work time, it is helpful to reduce the amount of samplings in comparison with some existing event-triggered mechanisms.

The remainder of this paper is organized as follows. Section II introduces the needed preliminaries. Section III presents our intermittent control scheme and develops the main quasi-synchronization criteria correspondingly. Section IV extends the proposed scheme to the event-triggered control. Section V demonstrates the effectiveness of our theoretical results by two simulation examples. Section VI concludes this paper.

Throughout this paper, $\mathbb{N}$ denotes the set of non-negative integers, $\mathbb{R}^n$ and $\mathbb{R}^{n \times m}$ are the n -dimensional Euclidean space and the set of all $n \times m$ real matrices, respectively. $x_\tau(t) = x(t - \tau(t))$ represents a continuous vector function with time-varying delay $\tau(t)$. $\mathcal{C}([-\tau_0, 0], \mathbb{R}^n)$ denotes the Banach space of continuous vector-valued functions mapping the interval $[-\tau_0, 0]$ into $\mathbb{R}^n$. For a given vector or matrix $H = (h_{ij})_{n \times m}$, the $|H|$ denotes the absolute value operation $|H| = (|h_{ij}|)_{n \times m}$, H^T represents the transpose of H . Further, if H is symmetric matrix, $H > 0$ ($H \geq 0$) means that H is positive definite (respectively, semidefinite), $\lambda_{\max}(H) > 0$ ($\lambda_{\min}(H) > 0$) is the maximum (respectively, minimum) eigenvalue of H . I and 0 denote identity matrix and zero matrix with appropriate dimensions, respectively. $\text{col}\{\cdots\}$ denotes a column vector, and $\text{diag}\{\cdots\}$ represents a block-diagonal matrix, respectively.

II. PRELIMINARIES

In this paper, we investigate the quasi-synchronization of delayed MNNs described by the following differential equation:

$$\dot{x}(t) = -Dx(t) + A(x(t))f(x(t)) + W(x_\tau(t))g(x_\tau(t)) + J \quad (1)$$

where $x(t) = \text{col}\{x_1(t), x_2(t), \dots, x_n(t)\}$ denotes the state variable of neurons; $J = \text{col}\{J_1, J_2, \dots, J_n\}$ is the external inputs; $\tau(t)$ denotes system delay satisfying $0 \leq \tau(t) \leq \tau_0$ and $\dot{\tau}(t) \leq \mu < 1$ (τ_0 and μ are two constants); $f(x(t)) = \text{col}\{f_1(x_1(t)), f_2(x_2(t)), \dots, f_n(x_n(t))\}$ and $g(x_\tau(t)) = \text{col}\{g_1(x_1(t - \tau(t))), g_2(x_2(t - \tau(t))), \dots, g_n(x_n(t - \tau(t)))\}$ denote the continuous activation functions; $D = \text{diag}\{d_1, d_2, \dots, d_n\}$ is a diagonal matrix with positive entries; $A(x(t)) = (a_{ij}(x_j(t)))_{n \times n}$ and $W(x_\tau(t)) = (w_{ij}(x_j(t - \tau(t))))_{n \times n}$ are the memristive synaptic weight matrices with

$$a_{ij}(x_j(t)) = \frac{\mathbf{M}_{aij}}{\mathbf{C}_i} \times \text{sgn}_{ij}$$

$$w_{ij}(x_j(t - \tau(t))) = \frac{\mathbf{M}_{wij}}{\mathbf{C}_i} \times \text{sgn}_{ij}$$

where $\text{sgn}_{ij} = 1$, if $i \neq j$ holds, otherwise, -1 ; the weights $a_{ij}(x_j(t))$ and $w_{ij}(x_j(t - \tau(t)))$ denote the strengths of the j th unit on the i th unit at time t and time $t - \tau(t)$, respectively;

$\mathbf{M}_{aij}$ and $\mathbf{M}_{wij}$ are the memductance of memristors $\mathbf{R}_{aij}$ and $\mathbf{R}_{wij}$. Here, $\mathbf{R}_{aij}$ stands for the memristor between neuron activation function $f_j(x_j(t))$ and $x_j(t)$, and $\mathbf{R}_{wij}$ stands for the memristor between neuron activation function $g_j(x_j(t - \tau(t)))$ and $x_j(t - \tau(t))$, respectively.

Note that $\mathbf{C}_i$ is changeless, and memductances $\mathbf{M}_{aij}$ and $\mathbf{M}_{wij}$ respond to change in the pinched hysteresis loops. Thus, $a_{ij}(x_j(t))$ and $w_{ij}(x_j(t - \tau(t)))$ will change as the change of pinched hysteresis loops. Based on the feature of memristor and the current-voltage characteristic, following [30]–[34], the definitions of $a_{ij}(x_j(t))$ and $w_{ij}(x_j(t - \tau(t)))$ are assumed to be:

$$a_{ij}(x_j(t)) = \begin{cases} a_{ij}^*, & |x_j(t)| \leq \mathcal{T}_j \\ a_{ij}^{**}, & |x_j(t)| > \mathcal{T}_j \end{cases}$$

$$w_{ij}(x_j(t - \tau(t))) = \begin{cases} w_{ij}^*, & |x_j(t - \tau(t))| \leq \mathcal{T}_j \\ w_{ij}^{**}, & |x_j(t - \tau(t))| > \mathcal{T}_j \end{cases}$$

where switching jumps $\mathcal{T}_j > 0$, a_{ij}^* , a_{ij}^{**} , w_{ij}^* , w_{ij}^{**} are constants. Throughout this paper, we denote $A^* = (a_{ij}^*)_{n \times n}$, $A^{**} = (a_{ij}^{**})_{n \times n}$, $W^* = (w_{ij}^*)_{n \times n}$, and $W^{**} = (w_{ij}^{**})_{n \times n}$, respectively.

The initial condition of MNNs (1) is defined as $x(s) = \varphi(s)$, $s \in [-\tau_0, 0]$, and $\varphi(s) \in \mathcal{C}([-\tau_0, 0], \mathbb{R}^n)$. It is customary to confine the functions f_j and g_j to the following condition.

Assumption (A₁): The neuron activation functions f_j and g_j are bounded and satisfy Lipschitz condition, namely, there exist positive scalars F_j , G_j , M_j^f , and M_j^g such that

$$\begin{aligned} |f_j(s_1) - f_j(s_2)| &\leq F_j |s_1 - s_2|, \quad |f_j(s_1)| \leq M_j^f \\ |g_j(s_1) - g_j(s_2)| &\leq G_j |s_1 - s_2|, \quad |g_j(s_1)| \leq M_j^g \end{aligned} \quad (2)$$

for any $s_1, s_2 \in \mathbb{R}$, $s_1 \neq s_2$.

Taking MNNs (1) as the master system, the associated slave system is described by the following equation:

$$\begin{aligned} \dot{y}(t) &= -Dy(t) + A(y(t))f(y(t)) + W(y_\tau(t))g(y_\tau(t)) \\ &\quad + J + u(t) \end{aligned} \quad (3)$$

where $u(t)$ is the intermittent controller, $A(y(t)) = (a_{ij}(y_j(t)))_{n \times n}$ and $W(y_\tau(t)) = (w_{ij}(y_j(t - \tau(t))))_{n \times n}$ with

$$a_{ij}(y_j(t)) = \begin{cases} a_{ij}^*, & |y_j(t)| \leq \mathcal{T}_j \\ a_{ij}^{**}, & |y_j(t)| > \mathcal{T}_j \end{cases}$$

$$w_{ij}(y_j(t - \tau(t))) = \begin{cases} w_{ij}^*, & |y_j(t - \tau(t))| \leq \mathcal{T}_j \\ w_{ij}^{**}, & |y_j(t - \tau(t))| > \mathcal{T}_j. \end{cases}$$

The initial condition of slave system (3) is defined as $y(s) = \psi(s)$, $s \in [-\tau_0, 0]$, and $\psi(s) \in \mathcal{C}([-\tau_0, 0], \mathbb{R}^n)$.

The systems (1) and (3) are differential equations with discontinuous right-hand sides, their solutions can be interpreted in the sense of Filippov. Since this issue has been fully considered in many papers, such as [28]–[31], and it does not play a significant role in the subsequent synchronization analysis, we here do not present more details on this issue.

In this paper, we also make the following assumption.

Assumption (A₂): There is no communication delay in the transmission among sensor, controller, and actuator.

Under Assumptions (A₁) and (A₂), the main aims of this paper are: 1) to devise a region-partitioning-dependent

intermittent control scheme and develop the associated quasi-synchronization criteria for MNNs (1) and (3), and 2) to extend the proposed intermittent scheme by taking into account the event-triggered control. At the end of this section, we introduce the following definition.

Definition 1 [5]: The MNNs (1) and (3) are said to be quasi-synchronized with an error level $l > 0$, if there exists a compact set $\mathcal{D}$, such that for any $\varphi(s), \psi(s) \in \mathcal{C}([-\tau, 0], \mathbb{R}^n)$, the error signal $e(t) = y(t) - x(t)$ converges to $\mathcal{D} := \{e \in \mathbb{R}^n : \|e(t)\| \leq l\}$ as $t \rightarrow \infty$.

III. REGION-PARTITIONING-DEPENDENT INTERMITTENT CONTROL

In this section, we first display our control scheme in details. Whereafter, we develop the quasi-synchronization criteria and present some necessary discussions.

A. Control Scheme

Let $e(t) = y(t) - x(t)$ be the synchronization error between MNNs (1) and (3). In this section, the controller $u(t)$ is designed as

$$u(t) = \begin{cases} Ke(t), & V(t) \in \mathcal{R}_1(t) \\ 0, & V(t) \in \mathcal{R}_2(t) \\ u(t^-), & V(t) \in \mathcal{R}_3(t) \end{cases} \quad (4)$$

where K is the control gain, $\mathcal{R}_i(t)$, $i = 1, 2, 3$, represent the partitions of non-negative real region $\mathbb{R}_+$ and satisfy $\bigcup_{i=1}^3 \mathcal{R}_i(t) = \mathbb{R}_+$ and $\mathcal{R}_i(t) \cap \mathcal{R}_j(t) = \emptyset$, $i \neq j$, $i, j \in \{1, 2, 3\}$. $V(t)$ is a given continuous positive definite function with respect to error $e(t)$, which is utilized to decide the work time and rest time of controller $u(t)$. t is the current time while t^- represents the previous time. Particularly, when $V(t_0) \in \mathcal{R}_3(t_0)$, we assume $u(t_0^-) = 0$.

From (4), the error dynamical system between MNNs (1) and (3) is recast into a switching system as follows:

$$\begin{cases} \dot{e}(t) = (-D + K)e(t) + F_e(t), & V(t) \in \mathcal{R}_1(t) & (5a) \\ \dot{e}(t) = -De(t) + F_e(t), & V(t) \in \mathcal{R}_2(t) & (5b) \\ \dot{e}(t) = -De(t) + F_e(t) + u(t^-), & V(t) \in \mathcal{R}_3(t) & (5c) \end{cases}$$

where

$$\begin{aligned} F_e(t) &= A(y(t))f(y(t)) - A(x(t))f(x(t)) \\ &\quad + W(y_\tau(t))g(y_\tau(t)) - W(x_\tau(t))g(x_\tau(t)). \end{aligned}$$

In essence, the intermittent controller (4) is a switching between $u(t) = Ke(t)$ and $u(t) \equiv 0$. Based on (4) and (5), the intermittent principle is demonstrated as follows.

- 1) When $V(t) \in \mathcal{R}_1(t)$, the controller $u(t) = Ke(t)$ is active and subsystem (5a) is working.
- 2) When $V(t) \in \mathcal{R}_2(t)$, the controller $u(t) \equiv 0$ is active and subsystem (5b) is working.
- 3) When $V(t) \in \mathcal{R}_3(t)$ and subsystem (5a) is active at time t^- , the controller $u(t) = Ke(t)$ is active and subsystem (5a) is working at time t .
- 4) When $V(t) \in \mathcal{R}_3(t)$ and subsystem (5b) is active at time t^- , the controller $u(t) \equiv 0$ is active and subsystem (5b) is working at time t .

Clearly, the time under cases 1 and 3 above belongs to work time, and the time under cases 2 and 4 above is rest time, respectively.

Remark 1: The subregions $\mathcal{R}_1(t)$, $\mathcal{R}_2(t)$, and $\mathcal{R}_3(t)$ represent the partitions of $\mathbb{R}_+$. Here, $\mathcal{R}_3(t)$ is taken as a transition region and is not empty. Moreover, it should be located in the middle of $\mathcal{R}_1(t)$ and $\mathcal{R}_2(t)$. The aim is to avoid the chattering of controller. Specially, if taking $\mathcal{R}_3(t) = \emptyset$, one has $\mathcal{R}_1(t) \cup \mathcal{R}_2(t) = \mathbb{R}_+$, and the controller (4) is changed as

$$u(t) = \begin{cases} Ke(t), & V(t) \in \mathcal{R}_1(t) \\ 0, & V(t) \in \mathcal{R}_2(t). \end{cases}$$

Let $\mathcal{B}(t)$ be the boundary between $\mathcal{R}_1(t)$ and $\mathcal{R}_2(t)$. Under the controller above, once the trajectory of $V(t)$ arrives at $\mathcal{B}(t)$, the controller may switch rapidly between $u(t) = Ke(t)$ and $u(t) \equiv 0$, which leads to the unwelcome chattering phenomenon. Such chattering usually causes the excessive equipment wear and consequently shorten their service life in practice. In contrast, the chattering will be eliminated if the transition region $\mathcal{R}_3(t)$ is chosen well.

The choice of $\mathcal{R}_i(t)$, $i = 1, 2, 3$, plays a crucial role in the processing of synchronization analysis. In this paper, we take the definitions of $\mathcal{R}_i(t)$ as follows:

$$\begin{aligned} \mathcal{R}_1(t) &= \{r \in \mathbb{R}_+ : r \geq a_1 + b_1 V(t_0) e^{-\lambda_1(t-t_0)}\} \\ \mathcal{R}_2(t) &= \{r \in \mathbb{R}_+ : r < a_2 + b_2 V(t_0) e^{-\lambda_2(t-t_0)}\} \\ \mathcal{R}_3(t) &= \mathbb{R}_+ \setminus (\mathcal{R}_1(t) \cup \mathcal{R}_2(t)). \end{aligned} \quad (6)$$

where $a_i \geq 0$, $b_i > 0$, and $\lambda_i > 0$, $i = 1, 2$, are given constants. Clearly, the boundaries between $\mathcal{R}_1(t)$ and $\mathcal{R}_3(t)$, $\mathcal{R}_2(t)$ and $\mathcal{R}_3(t)$, are formed naturally as follows, respectively:

$$\begin{aligned} \mathcal{B}_1(t) &= \{r \in \mathbb{R}_+ : r = a_1 + b_1 V(t_0) e^{-\lambda_1(t-t_0)}\} \\ \mathcal{B}_2(t) &= \{r \in \mathbb{R}_+ : r = a_2 + b_2 V(t_0) e^{-\lambda_2(t-t_0)}\}. \end{aligned}$$

In order to ensure the conditions $\mathcal{R}_i(t) \cap \mathcal{R}_j(t) = \emptyset$, $i \neq j$, $i, j \in \{1, 2, 3\}$, and $\mathcal{R}_3(t) \neq \emptyset$, the following condition should be ensured strictly for all $t \geq t_0$:

$$a_1 + b_1 V(t_0) e^{-\lambda_1(t-t_0)} > a_2 + b_2 V(t_0) e^{-\lambda_2(t-t_0)}. \quad (7)$$

In fact, there are many versions meeting the condition (7). For example: 1) $a_1 > a_2$, $b_1 > b_2$, and $\lambda_1 \leq \lambda_2$; 2) $a_1 > a_2 + b_2$; and 3) $a_1 > a_2$, $b_1 = b_2$, and $\lambda_1 < \lambda_2$, etc.

B. Main Results and Discussions

Before starting the quasi-synchronization criteria of this section, it is convenient to introduce the following lemma and proposition.

Lemma 1: Let $A = (a_{ij})_{n \times n}$ with $a_{ij} = \max\{|a_{ij}^*|, |a_{ij}^{**}|\}$, and $W = (w_{ij})_{n \times n}$ with $w_{ij} = \max\{|w_{ij}^*|, |w_{ij}^{**}|\}$. Then, under Assumption (A₁), the following inequalities hold:

$$\begin{aligned} |A(y(t))f(y(t)) - A(x(t))f(x(t))| &\leq AF|e(t)| + \Gamma_1 \\ |W(y_\tau(t))f(y_\tau(t)) - W(x_\tau(t))f(x_\tau(t))| &\leq WG|e_\tau(t)| + \Gamma_2 \end{aligned}$$

where $\Gamma_1 = |A^* - A^{**}|\gamma_1$ and $\Gamma_2 = |W^* - W^{**}|\gamma_2$, $\gamma_1 = \text{col}\{\gamma_{11}, \gamma_{12}, \dots, \gamma_{1n}\}$, $\gamma_2 = \text{col}\{\gamma_{21}, \gamma_{22}, \dots, \gamma_{2n}\}$,

$$\gamma_{1j} = \min\{F_j \mathcal{T}_j + |f_j(0)|, M_j^f\}, \gamma_{2j} = \min\{G_j \mathcal{T}_j + |g_j(0)|, M_j^g\} \quad j = 1, 2, \dots, n.$$

Proof: Note the boundedness of $f_j(\cdot)$ and $g_j(\cdot)$, the proof can be completed by the similar techniques in [25], [35], and [40]. Thus, the details are omitted here. ■

Proposition 1: For the chosen function $V(t)$ with initial condition $V(t_0)$, assume that the upper right-hand derivative of $V(t)$ along the solution of subsystem (5a) exists, and it satisfies the condition as follows:

$$D^+V(t) \leq -\alpha V(t) + c \quad (8)$$

where $\alpha > 0$ and $c \geq 0$ are constants, and $\alpha > \lambda_1$, $\alpha > \lambda_2$, $a_2 \geq (c/\alpha)$. Then, under the intermittent control scheme (4), (6), (7), there exists t^* such that, for all $t \geq t^*$, the following inequality holds:

$$V(t) \leq a_1 + b_1 V(t_0) e^{-\lambda_1(t-t_0)}. \quad (9)$$

Proof: Denote $\Omega(t) = \mathcal{B}_1(t) \cup \mathcal{R}_2(t) \cup \mathcal{R}_3(t)$. The inequality (9) is equivalent to that $V(t) \in \Omega(t)$ for all $t \geq t^*$. The proof can be completed by showing the following two facts.

- 1) If $V(t_0) \notin \Omega(t_0)$, there is a time t^* such that $V(t) \in \Omega(t)$ at $t = t^*$.
- 2) If there is a time t^* (t^* may be equal to t_0) such that $V(t^*) \in \Omega(t^*)$, the trajectory of $V(t)$ will dwell in region $\Omega(t)$ for $t > t^*$.

Fact 1: If $V(t_0) \notin \Omega(t_0)$, the subsystem (5a) is activated at $t = t_0$. By integrating (8) from t_0 to t , we obtain

$$V(t) \leq \frac{c}{\alpha} + \left(V(t_0) - \frac{c}{\alpha}\right) e^{-\alpha(t-t_0)}. \quad (10)$$

Let

$$H(t) = \frac{c}{\alpha} + \left(V(t_0) - \frac{c}{\alpha}\right) e^{-\alpha(t-t_0)} - a_1 - b_1 V(t_0) e^{-\lambda_1(t-t_0)}.$$

Since $V(t_0) \notin \Omega(t_0)$, i.e., $V(t_0) > a_1 + b_1 V(t_0)$, one has $H(t_0) = (1 - b)V(t_0) - a_1 > 0$ and $\lim_{t \rightarrow +\infty} H(t) = (c/\alpha) - a_1 < 0$. On the other hand, we can verify that $H(t)$ is monotonically decreasing for $t \in [t_0, t_0 + (1/[\alpha - \lambda_1]) \ln[(\alpha V(t_0) - c)/(\lambda_1 b_1 V(t_0))]]$ while monotonically increasing for $t \in [t_0 + [1/(\alpha - \lambda_1)] \ln[(\alpha V(t_0) - c)/(\lambda_1 b_1 V(t_0))], +\infty)$. Thus, there exists a unique t^* such that $H(t^*) = 0$. Moreover t^* falls into interval $[t_0, t_0 + (1/[\alpha - \lambda_1]) \ln[(\alpha V(t_0) - c)/(\lambda_1 b_1 V(t_0))]]$. This means that the inequality (9) holds at $t = t^*$, t^* is the unique solution of equation $H(t) = 0$, $t \geq t_0$.

It should be pointed out that the subsystem (5a) is still active at $t = t^*$. Then, by the similar analysis above, there is a time t^{**} such that the trajectory of $V(t)$ arrives at the boundary $\mathcal{B}_2(t)$ at time t^{**} .

Fact 2: If there is a time t^* such that $V(t^*) \in \Omega(t^*)$, the inequality (9) holds at time t^* . We can assume that there is a smallest time $t = T_1 \geq t^*$ such that the trajectory of $V(t)$ hits the boundary $\mathcal{B}_1(t)$ at $t = T_1$; otherwise, the inequality (9) holds for all $t \geq t^*$. At $t = T_1$, the subsystem (5a) is activated. Similar to the analysis in Fact 1, there will be a time T_2 such that the trajectory of $V(t)$ passes through the boundary $\mathcal{B}_2(t)$ and goes into $\mathcal{R}_2(t)$ at $t = T_2$. At T_2 , the subsystem (5b) is activated. So we can further assume that there is a time $t = T_3 > T_2$ such that the trajectory of $V(t)$ hits the boundary $\mathcal{B}_1(t)$ at $t = T_3$; otherwise, the inequality (9) holds for all $t \geq T_2$.

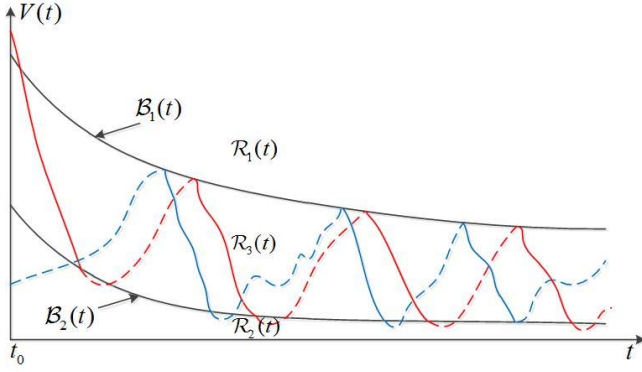


Fig. 1. Dynamical relationships among $V(t)$ with $V(t_0) \in \Omega(t_0)$ (blue line) [or $V(t)$ with $V(t_0) \notin \Omega(t_0)$ (red line), respectively], $B_1(t)$, and $B_2(t)$.

At T_3 , the subsystem (5a) is activated, then the trajectory of $V(t)$ will pass through the boundary $B_2(t)$ and go into $R_2(t)$ at T_4 . Repeating this procedure, one can assume that there are infinite switchings between subsystem (5a) and subsystem (5b) on $(t^*, +\infty)$.

Denote $\mathcal{T}_a = \cup_{k=0}^{+\infty} [T_{2s+1}, T_{2s+2})$ as the work time, i.e., the total activated time of subsystem (5a), and $\mathcal{T}_b = (t^*, T_1) \cup (\cup_{k=0}^{+\infty} [T_{2s+2}, T_{2s+3}))$ as the rest time, i.e., the total activated time of subsystem (5b), respectively.

When $t \in \mathcal{T}_b$, the trajectory of $V(t)$ falls into the region $R_2(t)$ or $R_3(t)$. Thus, inequality (9) holds naturally for $t \in \mathcal{T}_b$. When $t \in [T_{2s+1}, T_{2s+2}) \subset \mathcal{T}_a$, $s \in \mathbb{N}$, the subsystem (5a) is active. Moreover, at time T_{2s+1} , the trajectory of $V(t)$ is at the boundary of $B_1(t)$. Thus, we have

$$V(T_{2s+1}) = a_1 + b_1 V(t_0) e^{-\lambda_1 (T_{2s+1} - t_0)}. \quad (11)$$

From (8), we have

$$V(t) \leq \frac{c}{\alpha} + \left(V(T_{2s+1}) - \frac{c}{\alpha} \right) e^{-\alpha(t - T_{2s+1})} \quad t \in [T_{2s+1}, T_{2s+2}), \quad s \in \mathbb{N}. \quad (12)$$

Substituting (11) into (12), it yields

$$\begin{aligned} V(t) &\leq \frac{c}{\alpha} + \left(a_1 + b_1 V(t_0) e^{-\lambda_1 (T_{2s+1} - t_0)} - \frac{c}{\alpha} \right) e^{-\alpha(t - T_{2s+1})} \\ &= \frac{c}{\alpha} + \left(a_1 - \frac{c}{\alpha} \right) e^{-\alpha(t - T_{2s+1})} \\ &\quad + b_1 V(t_0) e^{-\lambda_1 (T_{2s+1} - t_0) - \alpha(t - T_{2s+1})} \\ &\leq a_1 + b_1 V(t_0) e^{\lambda_1 t_0 - \alpha t + (\alpha - \lambda_1) T_{2s+1}} \\ &\leq a_1 + b_1 V(t_0) e^{\lambda_1 t_0 - \alpha t + (\alpha - \lambda_1) t} \\ &= a_1 + b_1 V(t_0) e^{-\lambda_1 (t - t_0)}, \quad t \in [T_{2s+1}, T_{2s+2}), \quad s \in \mathbb{N}. \end{aligned} \quad (13)$$

The inequality (13) implies that inequality (9) holds for $t \in \mathcal{T}_a$. Note that $(t^*, +\infty) = \mathcal{T}_a \cup \mathcal{T}_b$, the proof is completed. ■

A graphical illustration on the proof of Proposition 1 is shown in Fig. 1.

- 1) If $V(t_0) \in \mathcal{R}_1(t_0)$, the subsystem (5a) is activated at t_0 . Since $a_2 > (c/\alpha)$ and the convergence rate of $V(t)$ is larger than the decay rates of boundaries $B_1(t)$ and $B_2(t)$ (i.e., $\alpha > \lambda_1$ and $\alpha > \lambda_2$), the trajectory of $V(t)$ will pass through the boundaries $B_1(t)$ and

$B_2(t)$, respectively, which activates the subsystem (5b). Whereafter, as time goes on, we can assume that the trajectory of $V(t)$ will touch the boundary $B_1(t)$ from the side of $\mathcal{R}_3(t)$. Otherwise, it is always below the line $a_1 + b_1 V(t_0) e^{-\lambda_1 (t - t_0)}$. When $V(t)$ touches the boundary $B_1(t)$, the subsystem (5a) is activated. Further, since $a_2 \geq (c/\alpha)$, $\alpha > \lambda_1$, and $\alpha > \lambda_2$, the trajectory of $V(t)$ will pass through $B_2(t)$ and go into $\mathcal{R}_2(t)$ again. This ensures that the trajectory of $V(t)$ does not exceed the boundary $B_1(t)$.

- 2) If $V(t_0) \in \mathcal{R}_2(t_0) \cup \mathcal{R}_3(t_0)$, the subsystem (5b) is activated at t_0 . It is assumed that there is a time such that the trajectory of $V(t)$ arrives at boundary $B_1(t)$ and the subsystem (5a) is activated first time. Then, similar to the description above, the trajectory of $V(t)$ will at most touch but not exceed the boundary $B_1(t)$.

Remark 2: In principle, the auxiliary positive function $V(t)$ can be chosen any one which satisfies the conditions in Proposition 1. However, for developing the quasi-synchronization criteria, one can take it as the form of Lyapunov function.

With Proposition 1 in mind, the remaining tasks are to choose a suitable auxiliary function $V(t)$ and estimate its upper right-hand derivative in the form of (9). Consider the following auxiliary function:

$$V(t) = e^T(t) P e(t) + \frac{1}{1 - \mu} \int_{t-\tau(t)}^t e^{\alpha(s-t)} e^T(s) Q e(s) ds \quad (14)$$

where $P, Q \in \mathbb{R}^{n \times n}$ are positive definite matrices. We next move forward to the quasi-synchronization analysis of MNNs.

Theorem 1: Under Assumptions (A_1) , (A_2) , and intermittent control scheme (4), (6), (7), (14), if there exist positive definite diagonal matrices $P, Q \in \mathbb{R}^{n \times n}$, such that, for the given control gain K , the following conditions hold:

$$a_2 \geq \frac{\Gamma_1^T \Gamma_1 + \Gamma_2^T \Gamma_2}{\alpha} \quad (15)$$

$$\Xi = \begin{bmatrix} \Xi_1 & PWG & P \\ GW^T P & -e^{-\alpha \tau_0} Q & 0 \\ P & 0 & -\frac{1}{2} I \end{bmatrix} < 0 \quad (16)$$

where $\Xi_1 = \alpha P + P(\tilde{K} - D + AF) + (\tilde{K} - D + AF)^T P + [1/(1 - \mu)]Q$, $\tilde{K} = (\tilde{k}_{ij})_{n \times n}$, and $\tilde{k}_{ii} = k_{ii}$, $\tilde{k}_{ij} = |k_{ij}|$ with $i \neq j$. Then the trajectory of error system (5) converges exponentially to the region $D := \{e \in \mathbb{R}^n : \|e(t)\| \leq \sqrt{(a_1/\lambda_{\min}(P))}\}$, i.e., the master system (1) and slave system (3) achieve quasi-synchronization with error level $\sqrt{(a_1/\lambda_{\min}(P))}$.

Proof: Calculating the upper right-hand derivative of $V(t)$ along the solution of subsystem (5a), it yields

$$\begin{aligned} D^+ V(t) &= -\alpha V(t) + \alpha e^T(t) P e(t) + 2e^T(t) P \dot{e}(t) \\ &\quad + \frac{1}{1 - \mu} e^T(t) Q e(t) - \frac{1 - \dot{\tau}(t)}{1 - \mu} e^{-\alpha \tau(t)} e^T(t) Q e_\tau(t) \\ &\leq \alpha |e^T(t) P e(t)| + 2|e^T(t) P [\tilde{K} - D] e(t)| \\ &\quad + |A(y(t))f(y(t)) - A(x(t))f(x(t))| \\ &\quad + |W(y_\tau(t))g(y_\tau(t)) - W(x_\tau(t))g(x_\tau(t))| \\ &\quad + \frac{1}{1 - \mu} |e^T(t) Q e(t)| - e^{-\alpha \tau_0} |e^T(t) Q e_\tau(t)|. \end{aligned}$$

Together with Lemma 1, the upper right-hand derivative of $V(t)$ can be estimated as

$$\begin{aligned} D^+V(t) \leq & -\alpha V(t) + |e^T(t)|(\alpha P + 2P(\tilde{K} - D))|e(t)| \\ & + 2|e^T(t)|P(AF|e(t)| + \Gamma_1) \\ & + 2|e^T(t)|P(WG|e_\tau(t)| + \Gamma_2) \\ & + \frac{1}{1-\mu}|e^T(t)|Q|e(t)| - e^{-\alpha\tau_0}|e^T(t)|Q|e_\tau(t)|. \end{aligned} \quad (17)$$

On the other hand, one has

$$2|e^T(t)|P\Gamma_1 \leq |e^T(t)|PP|e(t)| + \Gamma_1^T\Gamma_1 \quad (18)$$

$$2|e^T(t)|P\Gamma_2 \leq |e^T(t)|PP|e(t)| + \Gamma_2^T\Gamma_2. \quad (19)$$

Combing (17)–(19), it yields

$$D^+V(t) \leq -\alpha V(t) + \eta^T(t)\tilde{\Xi}\eta(t) + \Gamma_1^T\Gamma_1 + \Gamma_2^T\Gamma_2 \quad (20)$$

where $\eta(t) = \text{col}\{|e(t)|, |e_\tau(t)|\}$ and $\tilde{\Xi} = \begin{bmatrix} \tilde{\Xi}_1 & PWG \\ GW^TP & -e^{-\alpha\tau_0}Q \end{bmatrix}$ with $\tilde{\Xi}_1 = \alpha P + P(\tilde{K} - D + AF) + (\tilde{K} - D + AF)^TP + [1/(1-\mu)]Q + 2PP$.

Based on the well-known Schur complement lemma [46], the condition (16) is equivalent to $\tilde{\Xi} < 0$. Thus, from (20), we obtain

$$D^+V(t) \leq -\alpha V(t) + \Gamma_1^T\Gamma_1 + \Gamma_2^T\Gamma_2.$$

According to the Proposition 1, there exists a time t^* such that the following inequality holds for $t \geq t^*$:

$$V(t) \leq a_1 + b_1 V(t_0)e^{-\lambda_1(t-t_0)}.$$

Recalling the fact that $V(t) \geq e^T(t)Pe(t)$, the proof is completed. ■

In Theorem 1, we have developed the sufficient conditions which ensure the boundedness of synchronization error. In applications, we always desire that the synchronization error will converge to a given level. With this consideration, we are in a position to provide a co-design procedure of P , Q , and K , which ensures $\|e(t)\| \leq \sqrt{[a_1/(\lambda_{\min}(P))]} \leq l$ as $t \rightarrow +\infty$, $l > 0$ is a desired error level.

Step 1: Initialize the system parameters, including μ , τ_0 , F , G , A , W , Γ_1 , and Γ_2 . Choose the parameters a_i , b_i , λ_i ($i = 1, 2$), and α under conditions (7) and (15).

Step 2: Supplement a constraint condition as follows:

$$P \geq \frac{a_1}{\rho^2}I. \quad (21)$$

Step 3: Choose the control gain K .

Step 4: Check the conditions (16) and (21) by MATLAB. If yes, terminate the program and output the solutions of P , Q , and control gain K ; otherwise, go to back step 3.

Remark 3: Periodically intermittent control is a systematic control scheme, where the controller is devised as [30] and [39]–[42]

$$u(t) = \begin{cases} Ke(t), & mT \leq t < mT + \theta \\ 0, & mT + \theta \leq t < (m+1)T \end{cases} \quad (22)$$

where T denotes the control period, and $\theta \in (0, T]$ is the control width. Compared with scheme (22), our scheme has the following conspicuous differences.

- 1) Clearly, the control scheme (22) is clock-dependent, and the control width is periodic. However, in our scheme (4), (6), (7), (14), the control width depends upon the dynamical relationships among $\mathcal{R}_i(t)$ and $V(t)$. Then it is usually aperiodic.
- 2) For scheme (22), the index δ_w is usually computed as $\delta_w = (\theta/T)$, which depends upon the system parameters and some analytical techniques, such as the well-known mathematical induction (see [30], [40]). However, in this paper, the index δ_w is decided by the choice of $V(t)$, $\mathcal{R}_i(t)$, and the initial conditions.
- 3) When scheme (22) is considered, the function $V(t)$ is only viewed as the Lyapunov function. However, in this paper, the function $V(t)$ itself has been taken as an instrumental part of the control scheme. One of its roles is to determine the work time and rest time by monitoring its relationships with the subregions $\mathcal{R}_i(t)$ online.
- 4) For scheme (22), it is difficult to estimate the exponential convergence rate of error system accurately due to the complex relationships among the system parameters, T and θ . However, in our scheme, it only relies on the adjustable parameter λ_1 with $0 < \lambda_1 < \alpha$. Usually, a smaller scalar λ_1 yields a slower convergence rate of error system.

Remark 4: As stated in Remark 1, $\mathcal{R}_3(t)$ is a transition region, the aim is to eliminate the chattering phenomenon. In fact, the switching frequency of controller can be tuned by adjusting the size of $\mathcal{R}_3(t)$. Specially, it can be decreased by increasing the distance between $\mathcal{B}_1(t)$ and $\mathcal{B}_2(t)$, for example increasing the values of $a_1 - a_2$, $b_1 - b_2$, and $\lambda_1 - \lambda_2$. Usually, a larger size of $\mathcal{R}_3(t)$ corresponds to a lower switching frequency of controller.

Remark 5: In [45], a novel intermittent control scheme is proposed, where the switching of controller is decided by the norm of system states. However, the method in [45] is difficult to apply to the dynamical systems with time-delays. In contrast, the switching of controller in this paper is based on relationships among the subregions $\mathcal{R}_i(t)$ and delay-dependent function $V(t)$. This method takes the $V(t)$ as a part of the control scheme, thus it is easy to circumvent the trouble brought by time-delays. On the other hand, the results in [45] only apply to the case that the closed-loop systems are completely exponentially stable. While, we require the bounded stability in this paper. As a result, we have to discuss carefully the relations among parameters a_1 , a_2 , and c . Particularly, in the case of $a_1 = a_2 = c = 0$, the results of Proposition 1 remains its feasibility, and then it can also be applied directly to complete stability of closed-loop systems. Thus, our results are more general, and can be applied to the systems with more complexity.

In [30] and [32]–[34], some complete synchronization criteria for MNNs (1) and (3) have been presented by using classical continuous feedback control. However, all of these papers are based on the following assumption.

Assumption (A'_j): The neuron activation functions f_j and g_j satisfy both $f_j(\pm T_j) = g_j(\pm T_j) = 0$ and condition (2).

It is worth noting that the MNNs (1) and (3) are continuous under Assumption (A₁'), since $\lim_{x \rightarrow \pm \mathcal{T}_j} a_{ij}(x)f_j(x) = \lim_{x \rightarrow \pm \mathcal{T}_j} w_{ij}(x)g_j(x) = 0$ [35], [40]. Under Assumption (A₁'), the analysis in [30] and [31] results in

$$\begin{aligned} |A(y(t))f(y(t)) - A(x(t))f(x(t))| &\leq AF|e(t)| \\ |W(y_\tau(t))f(y_\tau(t)) - W(x_\tau(t))f(x_\tau(t))| &\leq WG|e_\tau(t)|. \end{aligned}$$

In addition, the Proposition 1 remains valid in the case $a_1 = a_2 = c = 0$. In this situation, the conclusion (9) is degraded into $V(t) \leq b_1 V(t_0) e^{-\lambda_1(t-t_0)}$. Therefore, we have the following corollary.

Corollary 1: Under Assumptions (A₁'), (A₂), and intermittent control scheme (4), (6), (7), (14), the trajectory of error system (5) converges exponentially to origin, i.e., the master system (1) and slave system (3) are exponentially synchronized, if there exist positive definite diagonal matrices $P, Q \in \mathbb{R}^{n \times n}$ such that, for the given control gain K , the following condition holds:

$$\hat{\Xi} = \begin{bmatrix} \Xi_1 & PWG \\ GW^T P & -e^{-\alpha t_0} Q \end{bmatrix} < 0 \quad (23)$$

where Ξ_1 is defined in Theorem 1.

IV. INTERMITTENT EVENT-TRIGGERED CONTROL

Up to now, the intermittent controller (4) has been devised to analyze the quasi-synchronization of MNNs. In digitalized era, such controller is usually implemented by an embedded digital computers. In such implementation, the original plant, sensor, controller, and actuator are connected over a shared digital communication network [19], [22], [47]. Wherein, the sensor transmits the measurements in a discrete sampling way. Based on this background, many event-triggered communication mechanisms have been devised aiming at reducing the communication burden and the energy consumption. Next, the proposed intermittent control scheme in Section III is applied to the event-triggered control, and an IETM is developed to analyze the quasi-synchronization of MNNs (1) and (3).

Without loss of generality, we denote $[T_{2s+1}, T_{2s+2})$ as intervals of work time, and $[T_{2s+2}, T_{2s+3})$, $s \in \mathbb{N}$, as intervals of rest time, respectively. For $t \in [T_{2s+2}, T_{2s+3})$ the controller $u(t) \equiv 0$, and then the sensor does not need to send the samplings. For $t \in [T_{2s+1}, T_{2s+2})$, the controller is active, and the sensor needs to send the samplings based on a preset trigger mechanism. Dwell in work time, the controller $u(t)$ in this section is taken as follows:

$$\begin{aligned} u(t) &= e^{-\beta(t-t_{s,k})} K e(t_{s,k}) \\ t &\in [t_{s,k}, t_{s,k+1}) \subset [T_{2s+1}, T_{2s+2}) \end{aligned} \quad (24)$$

where $\beta \geq 0$ is a given scalar, the elements of increasing sequence $\{t_{s,k}\}_{k \in \mathbb{N}}$ are execution times at which the controller is computed and updated. The associated IETM is devised as follows:

$$\begin{aligned} t_{s,0} &= T_{2s+1} \\ t_{s,k+1} &= \inf\{t \geq t_{s,k} | t \in [T_{2s+1}, T_{2s+2}) \\ &\quad \text{and } r^T(t) S r(t) > \rho e^T(t) S e(t)\} \end{aligned} \quad (25)$$

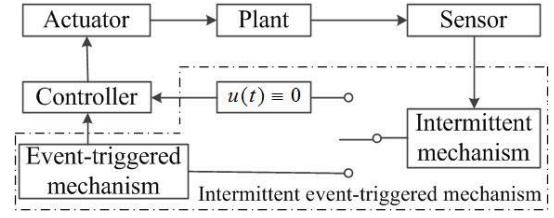


Fig. 2. Simple diagram for system with IETM. $u(t) \equiv 0$ means the rest time.

where $r(t) = e^{-\beta(t-t_{s,k})} e(t_{s,k}) - e(t)$, $S > 0$ is a diagonal matrix, ρ is a given threshold parameter.

Remark 6: The proposed IETM consists of an intermittent mechanism and an event-triggered mechanism. The former is utilized to decide the times T_{2s+1} and T_{2s+2} based on the intermittent principle in Section III. The later is applied to govern the time sequence $\{t_{s,k}\}_{k \in \mathbb{N}}$ by monitoring the condition $r^T(t) S r(t) > \rho e^T(t) S e(t)$. A system with IETM can be simply described by the diagram shown in Fig. 2. At time T_{2s+1} , the intermittent mechanism sends a signal to activate the event-triggered mechanism. Simultaneously, the event-triggered mechanism sends the samplings to controller such that the controller (24) is activated. At $t = T_{2s+2}$, no matter whether the condition $r^T(t) S r(t) > \rho e^T(t) S e(t)$ is violated or not, the intermittent mechanism will disconnect the connection with event-triggered mechanism and send a signal to activate the controller $u(t) \equiv 0$.

Remark 7: For $t \in [T_{2s+1}, T_{2s+2})$, the scheme (24), (25) can be viewed as a class of classical event-based state feedback control. By the similar technique in [49] and [53], one can show that such scheme eliminates the Zeno phenomenon. Thus, we omit the detailed discussion in this paper.

According to (24), the error system is converted into

$$\begin{cases} \dot{e}(t) = (-D + K)e(t) + F_e(t) + Kr(t) & V(t) \in \mathcal{R}_1(t) \\ \dot{e}(t) = -De(t) + F_e(t), & V(t) \in \mathcal{R}_2(t) \\ \dot{e}(t) = -De(t) + F_e(t) + u(t^-), & V(t) \in \mathcal{R}_3(t). \end{cases} \quad (26a) \quad (26b) \quad (26c)$$

Under controller (24), the Proposition 1 remains valid. Therefore, we have the following result.

Theorem 2: Under Assumptions (A₁), (A₂), and intermittent control scheme (6), (7), (14), (24), (25), if there exist positive definite diagonal matrices $P, Q, S \in \mathbb{R}^{n \times n}$, such that, for the given control gain K , inequality (15) and the following condition hold:

$$\Xi_{\text{event}} = \begin{bmatrix} \Xi_1 + \rho S & PWG & P|K| & P \\ GW^T P & -e^{-\alpha t_0} Q & 0 & 0 \\ |K^T|P & 0 & -S & 0 \\ P & 0 & 0 & -\frac{1}{2}I \end{bmatrix} < 0 \quad (27)$$

where Ξ_1 follows the same definition as that in Theorem 1. Then the trajectory of error system (5) converges exponentially to the region $D := \{e \in \mathbb{R}^n | \|e(t)\| \leq \sqrt{(a_1/\lambda_{\min}(P))}\}$, i.e., the master system (1) and slave system (3) achieve quasi-synchronization with error level $\sqrt{(a_1/\lambda_{\min}(P))}$.

Proof: Calculating the upper right-hand derivative of $V(t)$ along the solution of subsystem (26a), it yields

$$\begin{aligned} D^+V(t) + \alpha V(t) &\leq |e^T(t)|(\alpha P + 2P(\tilde{K} - D))|e(t)| + 2|e^T(t)|P|K||r(t)| \\ &\quad + 2|e^T(t)|P(AF|e(t)| + \Gamma_1) + 2|e^T(t)|P(WG|e_\tau(t)| + \Gamma_2) \\ &\quad + \frac{1}{1-\mu}|e^T(t)|Q|e(t)| - e^{-\alpha t_0}|e^T(t)|Q|e_\tau(t)|. \end{aligned} \quad (28)$$

From the IETM (25), on $[t_{s,k}, t_{s,k+1})$, one has

$$0 \leq \rho|e^T(t)|S|e(t)| - |r^T(t)|S|r(t)|. \quad (29)$$

Combining (28) and (29), we obtain

$$D^+V(t) \leq -\alpha V(t) + \eta^T(t)\tilde{\Xi}_{\text{event}}\eta(t) + \Gamma_1^T\Gamma_1 + \Gamma_2^T\Gamma_2 \quad (30)$$

where $\tilde{\eta}(t) = \text{col}\{|e(t)|, |e_\tau(t)|, |r(t)|\}$, $\tilde{\Xi}_{\text{event}} = \begin{bmatrix} \Xi_1 + \rho S & PWG & P|K| \\ GW^TP & -e^{-\alpha t_0}Q & 0 \\ |K^T|P & 0 & -S \end{bmatrix}$, and $\tilde{\Xi}_1$ follows the same definition in (20). By the similar analysis in the proof of Theorem 1, we complete the proof. ■

As a consequence of Theorem 2, we have the following corollary.

Corollary 2: Under Assumptions (A'_1) , (A_2) , and intermittent control scheme (6), (7), (14), (24), (25), the trajectory of error system (5) converges exponentially to origin, i.e., the master system (1) and slave system (3) are exponentially synchronized, if there exist positive definite diagonal matrices P , Q , $S \in \mathbb{R}^{n \times n}$, such that, for the given control gain K , the following condition holds:

$$\hat{\Xi}_{\text{event}} = \begin{bmatrix} \Xi_1 + \rho S & PWG & P|K| \\ GW^TP & -e^{-\alpha t_0}Q & 0 \\ |K^T|P & 0 & -S \end{bmatrix} < 0 \quad (31)$$

where Ξ_1 is defined in Theorem (1).

Remark 8: The event-triggered control has been fully investigated for the dynamical behaviors of line/nonlinear systems [19], [20], [48]–[51], including the memristive systems [52]–[54]. For example, the papers [52] and [53] first investigated the event-based quantized sampled control and event-based sliding-model control of memristive systems, respectively. Compared with the existing results, the main difference of IETM (25) lies in that it is only concerned with the events in work time $[T_{2s+1}, T_{2s+2})$ rather than $[t_0, +\infty)$. Meanwhile, for $t \in [T_{2s+2}, T_{2s+3})$, the controller $u(t) \equiv 0$, and then the sensor does not need to transmit the sampling signals, even though the condition $r^T(t)Sr(t) > \rho e^T(t)Se(t)$ is satisfied. Therefore, the devised IETM (25) can reduce the amount of samplings. To the best of our knowledge, such IETM has been rarely reported in the existing references.

V. NUMERICAL EXAMPLES

In this section, two numerical examples are presented to show the effectiveness of our control scheme. The first example comes from [30] and is utilized to exposit the advantages of our intermittent control scheme. The other task of this example is to demonstrate the validity of Corollary 2. The second example is applied to illustrate the effectiveness of

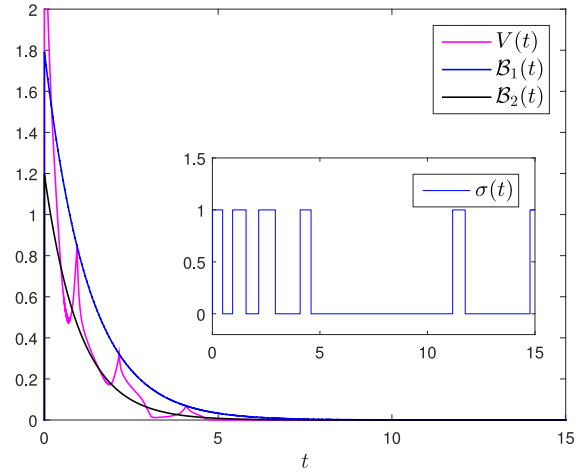


Fig. 3. Trajectories of $V(t)$, $B_1(t)$, $B_2(t)$, and $\sigma(t)$ in Example 1.

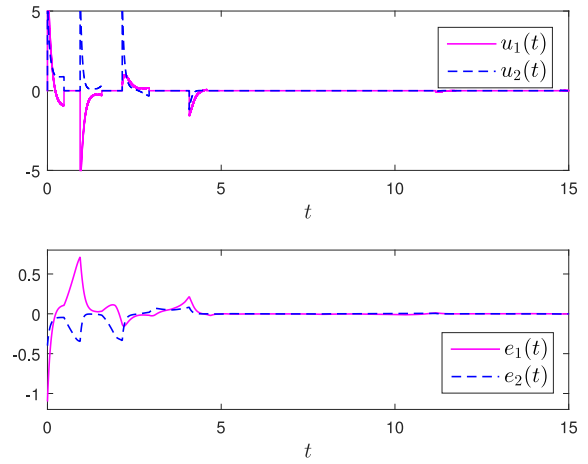


Fig. 4. Trajectories of intermittent controller $u(t)$ and synchronization error $e(t)$ in Example 1.

Theorems 1 and 2, respectively. To describe the switching between $u(t) = Ke(t)$ [or $u(t) = e^{-\beta(t-t_{s,k})}Ke(t_{s,k})$] and $u(t) \equiv 0$ clearly, we introduce a signal $\sigma(t)$ in the examples. Namely, if $\sigma(t) = 1$, then $u(t) = Ke(t)$ [or $u(t) = e^{-\beta(t-t_{s,k})}Ke(t_{s,k})$], otherwise, $u(t) \equiv 0$.

Example 1: Consider a 2-neuron master system (1) with the following memristive synaptic weights [30]:

$$\begin{aligned} d_1 &= 1, \quad d_2 = 1, \quad a_{11}(x_1(t)) = 1 \\ a_{22}(x_2(t)) &= 1.8 \\ a_{12}(x_2(t)) &= \begin{cases} 7, & |x_2(t)| \leq 1 \\ 5, & |x_2(t)| > 1 \end{cases} \\ a_{21}(x_1(t)) &= \begin{cases} 0.8, & |x_1(t)| \leq 1 \\ 1, & |x_1(t)| > 1 \end{cases} \\ w_{11}(x_1(t - \tau(t))) &= \begin{cases} -1.5, & |x_1(t - \tau(t))| \leq 1 \\ -1.2, & |x_1(t - \tau(t))| > 1 \end{cases} \\ w_{12}(x_2(t - \tau(t))) &= \begin{cases} 1.0, & |x_2(t - \tau(t))| \leq 1 \\ 0.8, & |x_2(t - \tau(t))| > 1 \end{cases} \\ w_{21}(x_1(t - \tau(t))) &= \begin{cases} 0.8, & |x_1(t - \tau(t))| \leq 1 \\ 1, & |x_1(t - \tau(t))| > 1 \end{cases} \\ w_{22}(x_2(t - \tau(t))) &= \begin{cases} -1.6, & |x_2(t - \tau(t))| \leq 1 \\ -1.4, & |x_2(t - \tau(t))| > 1. \end{cases} \end{aligned}$$

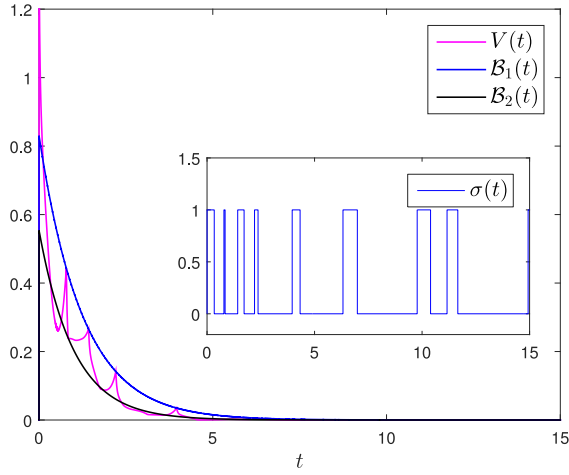


Fig. 5. Trajectories of $V(t)$, $B_1(t)$, $B_2(t)$, and $\sigma(t)$ in Example 1.

The associated slave system is defined in (3). The time-varying delay is taken as $\tau(t) = 0.75 - 0.25 \cos(t)$, and the activation functions are chosen as $f_1(s) = \sin(|s| - 1)$ and $f_2(s) = \tanh(|s| - 1)$, respectively. A straightforward computation gives $\tau_0 = 1$, $\mu = 0.25$, and $F = G = I$. The initial states of master system and slave system are taken as $x(t) = \text{col}\{0.8, -0.4\}$, and $y(t) = \text{col}\{-0.3, -0.8\}$, $t \in [-1, 0]$, respectively.

First, following [30], we take the control gain as $K = \begin{bmatrix} -8 & 2 \\ 1 & -17 \end{bmatrix}$. By solving the condition in Corollary 1 with $\alpha = 1.2$, we obtain

$$P = \text{diag}\{0.5773, 1.0029\}, \quad Q = \text{diag}\{2.2464, 5.9226\}.$$

Selecting $b_1 = 0.6$, $b_2 = 0.4$, $\lambda_1 = 0.8$, and $\lambda_2 = 1.0$, then all the conditions in Corollary 1 are satisfied. The simulation results are shown in Figs. 3 and 4. Fig. 3 reveals the relations among the trajectories of $V(t)$, $B_1(t)$, $B_2(t)$, and the corresponding signal $\sigma(t)$. Fig. 4 displays the trajectories of intermittent controller $u(t)$ and the associated synchronization error $e(t)$. It should be pointed out that, in this example, the index $\delta_w = 21.31\%$. However, it is 95.75% in [30]. Clearly, the proposed intermittent control scheme has achieved a fundamental improvement in comparison with [30].

Second, we will check the effectiveness of Corollary 2. We assume the threshold parameter $\rho = 0.2$ and scalar $\beta = 2.0$. Solving the condition (31) with $K = -17.5I$, it yields

$$P = \text{diag}\{0.2495, 0.6833\}, \quad Q = \text{diag}\{1.3855, 2.3851\} \\ S = \text{diag}\{9.8770, 26.1530\}.$$

The trajectories of $V(t)$, $B_1(t)$, and $B_2(t)$ are presented in Fig. 5. The trajectories of controller (24) and corresponding synchronization error $e(t)$ are depicted in Fig. 6. Under the event-based intermittent controller (24), the index $\delta_w = 20.51\%$, which is also much less than the one in [30]. Here, the amount of samplings is 83.

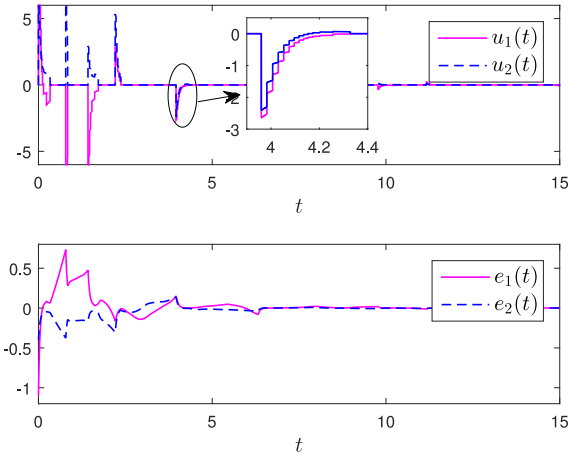


Fig. 6. Trajectories of intermittent controller $u(t)$ and synchronization error $e(t)$ in Example 1.

Example 2: Consider a 2-neuron master system (1) with the following memristive synaptic weights:

$$\begin{aligned} d_1 &= 1, \quad d_2 = 1, \quad a_{11}(x_1(t)) = a_{22}(x_2(t)) = 0 \\ w_{11}(x_1(t - \tau(t))) &= -1, \quad w_{22}(x_2(t - \tau(t))) = -0.2 \\ a_{12}(x_2(t)) &= \begin{cases} 3, & |x_2(t)| \leq 0.2 \\ 3.5, & |x_2(t)| > 0.2 \end{cases} \\ a_{21}(x_1(t)) &= \begin{cases} 4, & |x_1(t)| \leq 0.2 \\ 4.5, & |x_1(t)| > 0.2 \end{cases} \\ w_{12}(x_2(t - \tau(t))) &= \begin{cases} 2, & |x_2(t - \tau(t))| \leq 0.2 \\ 2.5, & |x_2(t - \tau(t))| > 0.2 \end{cases} \\ w_{21}(x_1(t - \tau(t))) &= \begin{cases} 3, & |x_1(t - \tau(t))| \leq 0.2 \\ 3.5, & |x_1(t - \tau(t))| > 0.2 \end{cases} \end{aligned}$$

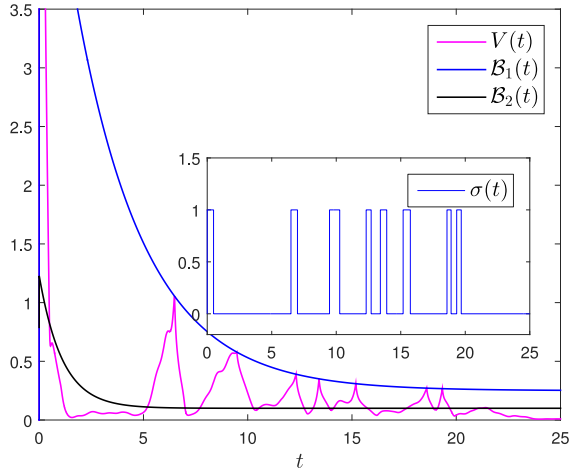
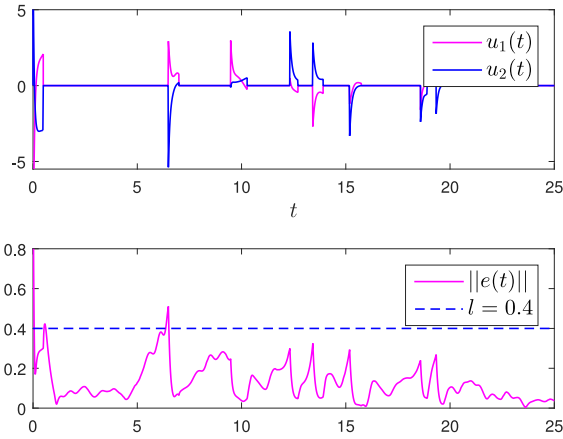
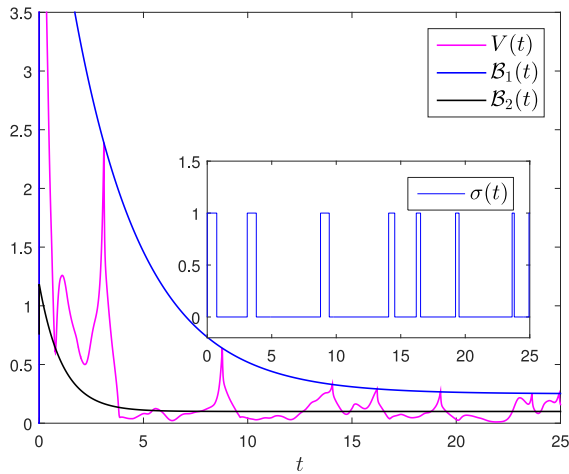
The associated slave system is defined in (3). We take $f_j(s) = g_j(s) = \sin(s)$, $j = 1, 2$, and $\tau(t) = 0.75 - 0.25 \sin(t)$, respectively. By simple computing, we get $\tau_0 = 1$, $\mu = 0.25$, $F = G = I$, and $\Gamma_1^T \Gamma_1 + \Gamma_2^T \Gamma_2 = 0.08$. For simulation, we assume that the desired error level is $l = 0.4$ and parameter $\alpha = 1.0$. Thus, from (2), one has $a_2 \geq 0.08$. In this example, we take $a_1 = 0.25$, $a_2 = 0.1$, $b_1 = 0.5$, $b_2 = 0.1$, $\lambda_1 = 0.3$, $\lambda_2 = 0.9$. According to condition (21), the matrix P should satisfy $P \geq (25/16)I$. By choosing $x(t) = \text{col}\{-0.6, 0.3\}$, and $y(t) = \text{col}\{0.5, -0.4\}$, $t \in [-1, 0]$, we next move to check the validity of the designed controllers (4) and (24), respectively.

First, we consider the controller (4). By taking control gain as $K = -12I$ and checking the condition (16), we obtain the following feasible solution:

$$P = \text{diag}\{1.6524, 1.5869\}, \quad Q = \text{diag}\{9.8575, 6.7866\}.$$

The simulation results are provided in Figs. 7 and 8. Fig. 7 presents the trajectories of $V(t)$, $B_1(t)$, $B_2(t)$, and switching signal $\sigma(t)$. Here, the index $\delta_w = 15.57\%$. Fig. 8 depicts the trajectories of intermittent controller $u(t)$ and synchronization error $\|e(t)\|$. From Fig. 8, we can find that our control scheme contributes to the quasi-synchronization with $\|e(t)\| \leq \sqrt{(0.25/1.5869)} < 0.4$.

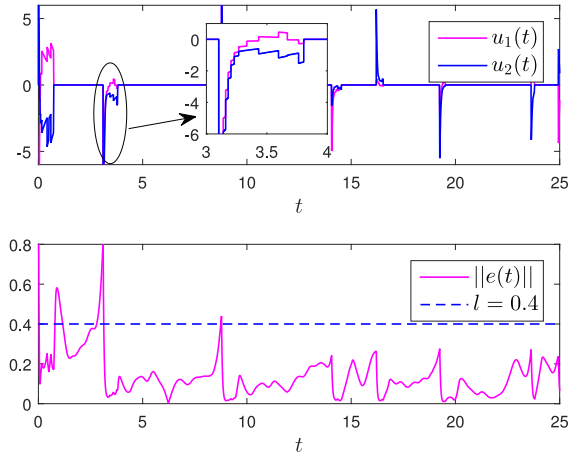
Second, let the threshold parameter $\rho = 0.2$ and $\beta = 1.5$, we consider the event-based intermittent controller (24). By taking control gain as $K = -21.7I$ and checking the

Fig. 7. Trajectories of $V(t)$, $B_1(t)$, $B_2(t)$, and $\sigma(t)$ in Example 2.Fig. 8. Trajectories of intermittent controller $u(t)$ and synchronization error $\|e(t)\|$ in Example 2.Fig. 9. Trajectories of $V(t)$, $B_1(t)$, $B_2(t)$, and $\sigma(t)$ in Example 2.

condition (16), we obtain the following feasible solution:

$$P = \text{diag}\{1.6449, 1.5664\}, \quad Q = \text{diag}\{9.7517, 6.6841\} \\ S = \text{diag}\{79.8200, 75.6836\}.$$

The trajectories of $V(t)$, $B_1(t)$, $B_2(t)$, and $\sigma(t)$ are plotted in Fig. 9, and the corresponding controller and synchronization

Fig. 10. Trajectories of intermittent controller $u(t)$ and synchronization error $\|e(t)\|$ in Example 2.

error $\|e(t)\|$ are shown in Fig. 10. In this case, the amount of samplings is 79, and the index $\delta_w = 13.68\%$.

VI. CONCLUSION

In this paper, we have made efforts toward exploring the quasi-synchronization analysis of MNNs by proposing the region-partitioning-dependent intermittent control scheme. This scheme decides whether the controller is imposed or not by monitoring the dynamical relationships among three sub-regions of non-negative real region and an auxiliary positive definite function, then it usually yields aperiodic work time and rest time. Under the framework of our scheme, sufficient conditions have been developed to guarantee the boundedness of synchronization error. In order to achieve a desired error level, a co-design procedure has been provided to determine the control gain and weighting matrices in the auxiliary function correspondingly. Together with the event-triggered control, the proposed intermittent scheme has been extended to devise an IETM aiming at reducing the communication load in the network. Since the proposed IETM only focuses on the events in work time, it is helpful to reduce the amount of samplings. Two simulation examples have been employed to demonstrate the feasibility of our control scheme.

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